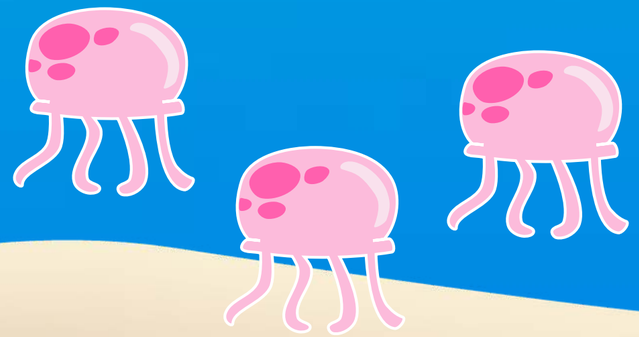
MODULE 3

MICROBIAL BIOTECHNOLOGY



EXPECTED LEARNING OUTCOME

1. Describe features of bacteria that make them useful for applications in biotechnology.
2. Provide examples of how yeasts are used in biotechnology.
3. List examples of medically important proteins that are produced in bacteria using recombinant DNA technology.
4. Explain how alcohol and lactic acid fermentation processes can be used to produce common foods and beverages.
5. Describe the role microorganisms play in the development and production of vaccines; provide examples of different vaccines.

STRUCTURE OF MICROBES

Eukaryotic

- true nucleus

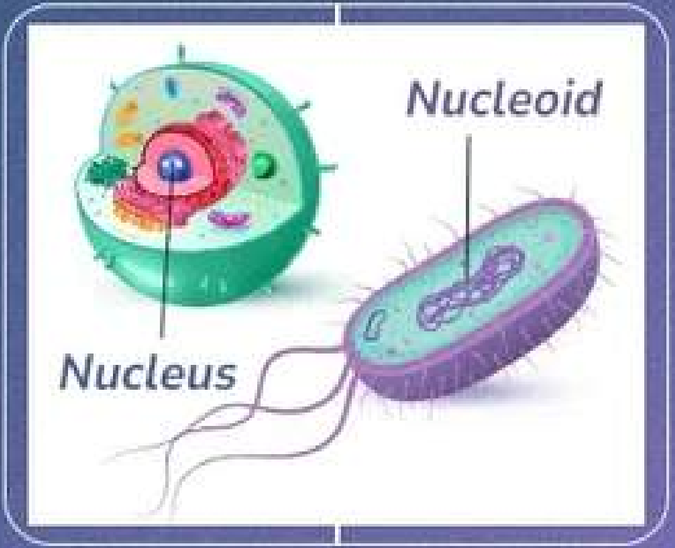
Prokaryotic

- lacks a true nucleus
- two major domains: **Bacteria** and **Archea**
- halophiles and thermophiles

Difference Between

Nucleus & Nucleoid

- Spherical or oval shaped
- Double membrane bound
- Contains DNA, RNA, histones etc



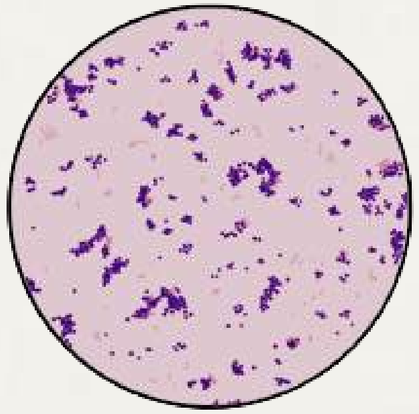
- Irregularly shaped in prokaryotes
- Lacks a membrane
- Contains mainly DNA

KEY STRUCTURAL FEATURES OF BACTERIA

- DNA is single circular chromosome
- Plasmid
- No membrane-bound organelles
- Cell wall is made of peptidoglycan
- Some have capsule

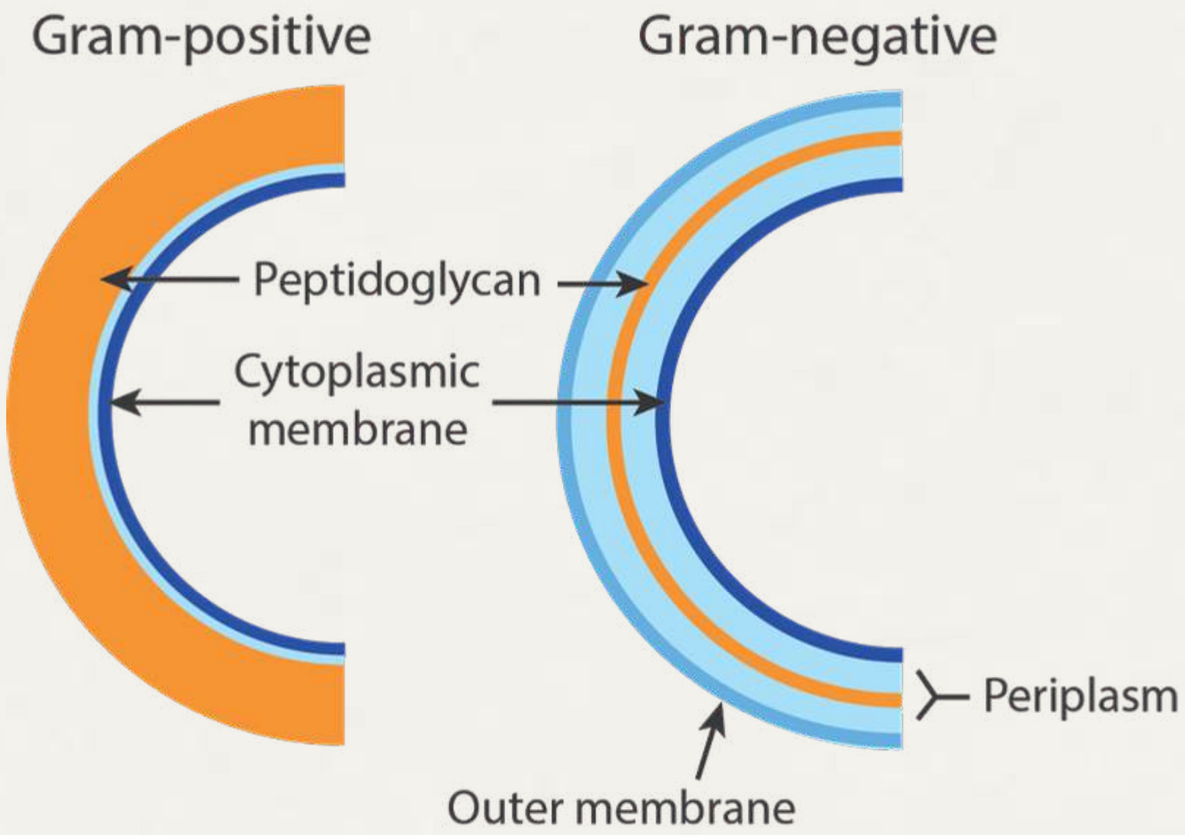
GRAM STAIN TECHNIQUE

- **Gram-positive bacteria**: purple and thick cell walls



- **Gram-negative bacteria**: pink and thin cell walls

GRAM STAIN TECHNIQUE



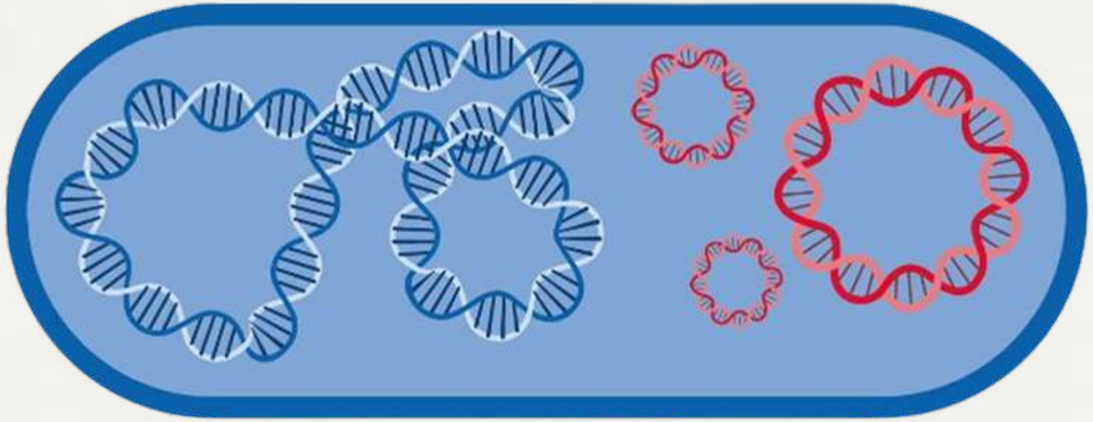
BACTERIAL SHAPES AND ARRANGEMENTS

COCCI

BACILLI

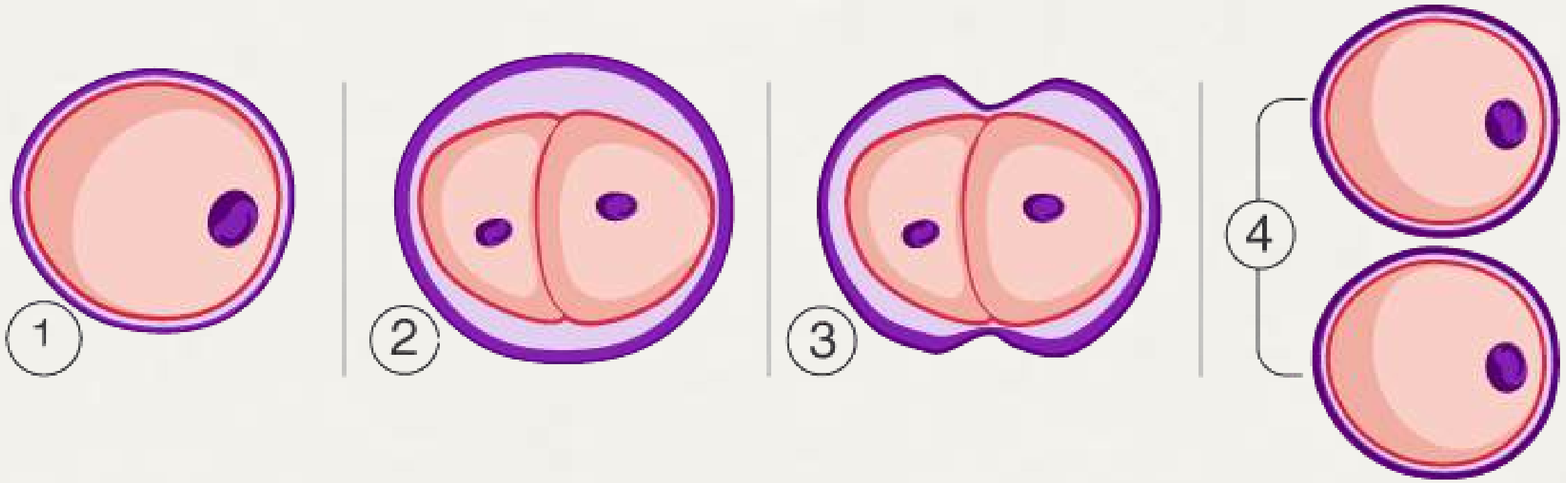
SPIRILLA

GENETIC MATERIAL AND PLASMID



- 2 to 4 million base pairs
- contain plasmids

GROWTH AND REPRODUCTION OF BACTERIA



Binary Fission

BACTERIA AS MODEL ORGANISMS

- grow quickly
- easy to culture
- genetically manipulated

YEASTS ARE IMPORTANT MICROBES

- Brewing beer
- Recombinant human antibodies
- Biotechnological products
- Model organism
- Yeast artificial chromosomes

STRUCTURE AND CHARACTERISTICS OF YEAST

- Vary in size and shape: **spherical**, **elliptical**, or **cylindrical**.
- Can grow **aerobic** and **anaerobic**

STRUCTURE AND CHARACTERISTICS OF YEAST

Saccharomyces cerevisiae

- first eukaryotic organism to have its entire genome sequenced
- gene expression closely resembles that in human cells

INDUSTRIAL AND BIOTECHNOLOGY USES OF YEAST

Pichia pastoris

- grows to high cell densities in liquid culture
- possesses strong promoters
- batch processing and large-scale production of proteins and enzymes
- used in the production of vaccines, enzymes, and recombinant proteins

MICROBIAL ENZYMES

- Microbial enzymes: speed up reactions
- Uses of microbial enzymes: food, molecular biology, detergents
- *E. coli*: Restriction enzyme and DNA polymerase
- Techniques: DNA labeling, gene cloning, and polymerase chain reaction (PCR)

Example: Taq polymerase (*Thermus aquatics*) and Pfu DNA polymerase (*Pyrococcus furiosus*)

- **Bioprospecting**- species survived up to 121°C

OTHER IMPORTANT ENZYMES

1. **Cellulase**

- *E. coli*
- breaks down cellulose
- digestible animal feed
- soften fabrics in textile
- *Trichoderma reesei and Aspergillus niger*

OTHER IMPORTANT ENZYMES

2. Subtilisin
- *Bacillus subtilis*
- remove stains in laundry detergents

3. Amylases
- break down starches
- corn syrup production

BACTERIAL TRANSFORMATION

Bacterial transformation is the process by which bacteria take in DNA from their surrounding environment.

MAKING BACTERIAL CELLS COMPETENT

treating bacteria with an ice-cold solution of calcium chloride

TRANSFORMATION PROCESS

1. Mix recombinant plasmid DNA with chilled competent cells.
2. Place on ice → DNA attaches to cell surface.
3. Apply heat shock (37°C–42°C, ~1 min) → DNA enters cell.
4. Allow cells to recover in nutrient broth → express new genes

ANTIBIOTIC SELECTION AND RECOMBINANT EXPRESSION

Only cells that have successfully taken up the plasmid DNA with the resistance gene will survive and form colonies. This process is called **antibiotic selection**.

The process is called **transformation** because the bacterial cells are genetically altered to gain new properties.

- *E. coli* transformed with the green fluorescent protein.

ELECTROPORATION

Electroporation is another method for introducing DNA into bacterial cells. In this technique, an instrument known as an electroporator delivers a brief electrical pulse to a mixture of bacterial cells and DNA.

USING MICROBES FOR EVERYDAY APPLICATION

Microbial biotechnology uses natural or genetically engineered microorganisms to produce essential products. These microbes play a vital role in food production and in manufacturing medically important recombinant proteins that improve human health.

USING MICROBES FOR EVERYDAY APPLICATION

Microorganisms are used in the production of foods such as bread, yogurt, cheese, beer, and wine.

Cheese production relies on the enzyme **rennin,** also called **chymosin**, which coagulates milk; the rennin gene was cloned and expressed in *Aspergillus niger*, becoming the first genetically engineered food ingredient approved by the FDA in 1990.

USING MICROBES FOR EVERYDAY APPLICATION

Lactic acid bacteria such as *Lactococcus lactis* and *Lactobacillus acidophilus* ferment milk by converting lactose into lactic acid, giving cheese and yogurt their flavor and texture.

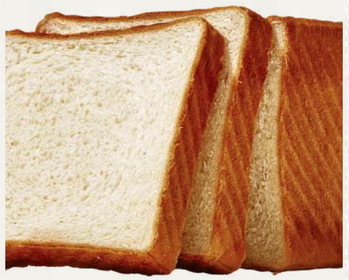
FERMENTING MICROBES

Fermentation is a key microbial process in which microbes convert sugars into useful products such as lactic acid, ethanol, and carbon dioxide.d texture.

Lactobacillus bulgaricus and S*treptococcus thermophilus*- lactic acid fermentation

FERMENTING MICROBES

Yeasts like *Saccharomyces cerevisiae* are used in brewing beer, making wine, and baking bread.



Lactic acid bacteria such as *Streptococcus thermophilus* and *Lactobacillus bulgaricus* produce yogurt and sour cream.

PRODUCTION OF RECOMBINANT INSULIN IN BACTERIA

Insulin is a hormone made by the beta cells of the pancreas and regulates blood sugar levels.

When the body cannot produce insulin, a person develops **Type I diabetes mellitus**, which can cause high blood sugar, high blood pressure, poor blood circulation and nerve damage.

PRODUCTION OF RECOMBINANT INSULIN IN BACTERIA

Before Recombinant DNA Technology

- Insulin comes from pig and cow
- It was expensive, inefficient, and caused allergic reactions.

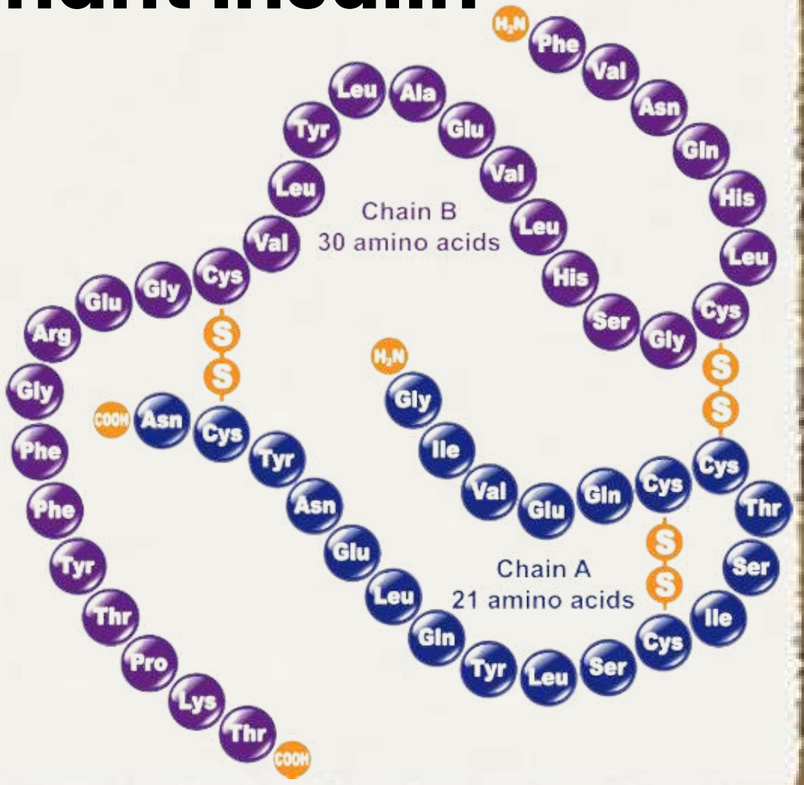
After Recombinant DNA Technology

- humulin into bacteria

PRODUCTION OF RECOMBINANT INSULIN IN BACTERIA

How Bacteria Produce Recombinant Insulin

Human insulin consists of A chain (21 amino acids)and B chain (30 amino acids)



PRODUCTION OF RECOMBINANT INSULIN IN BACTERIA

In humans, insulin is made as one molecule and later folded into its active form. Meanwhile, in bacteria, A and B chain genes are cloned separately into plasmids and linked to the lacZ gene.

1. Bacteria produce a fusion protein (β-gal + insulin), which is then isolated.
2. Insulin is separated from β-gal, and the purified A and B chains are combined to form active insulin.
3. Recombinant insulin is purified for medical use, making production faster, safer, and cheaper for diabetic patients.

OTHER THERAPEUTIC PROTEINS FROM BACTERIA

- **Human growth hormone** - for treating growth deficiencies.
- **Interferons and interleukins** - for treating immune-related diseases.
- **Clotting factors** - for hemophilia.
- **Erythropoietin** (EPO) - for stimulating red blood cell production.

DISCOVERY OF ANTIBIOTICS

Microbes produce antibiotics, which are substances that inhibit or kill other microbes.

The discovery of penicillin by Alexander Fleming in 1928, when *Penicillium notatum* killed nearby *Staphylococcus aureus*, marked the start of the antibiotic era and led to the first widely used antibiotic in humans.

HOW ANTIBIOTIC WORKS

Antibiotics kill or stop bacteria by targeting essential bacterial processes such as:

- Cell wall synthesis (e.g., Penicillin)
- Protein synthesis (ribosome inhibitors)
- DNA replication
- Metabolic enzyme activity

PROBLEM OF ANTIBIOTIC RESISTANCE

Overuse and misuse of antibiotics in humans and animals have led to antibiotic-resistant bacteria

Examples include resistant forms of:

- *Staphylococcus aureus*
- *Pseudomonas aeruginosa*
- *Streptococcus pneumoniae*
- *Mycobacterium tuberculosis*

NEW FRONTIERS IN ANTIMICROBIAL RESEARCH

1. **Recombinant bacteria that prevent disease** – *Streptococcus mutans* engineered to reduce tooth decay.
2. **Natural compounds from unusual sources**– Licorice-based lollipops (UCLA) kill *S. mutans*.

VACCINES

Vaccines prepare the immune system to recognize and fight microbes before infection.

Edward Jenner (1796) used cowpox to protect against smallpox, leading to vaccination; smallpox was eradicated in 1980.

VACCINES

Examples of vaccines:

- **DPT** (diphtheria, pertussis, tetanus) – bacterial toxins
- **MMR** (measles, mumps, rubella) – viral infections
- **Polio vaccines** (Salk & Sabin) – reduced poliomyelitis cases

VACCINES

Modern biotechnology: Recombinant DNA allows bacteria to produce antigens for vaccines safely.

- *E. coli* produces hepatitis B surface antigen
- Recombinant bacteria also produce antigens for HPV and other viral vaccines

PRIMER ON ANTIBODIES

The immune system protects the body by recognizing antigens (microbes, toxins, or microbial parts).

- B cells produce antibodies that attach to specific antigens, with help from T cells; some become plasma cells that release antibodies into body fluids.
- Some B cells become memory cells, allowing faster response upon future exposure—this is the basis of vaccination.

PRIMER ON ANTIBODIES

- Vaccines stimulate antibody production so the immune system can respond quickly to real infections.
- Biotechnology enhances immunity by producing recombinant antibodies and vaccines for diseases such as hepatitis, influenza, and HIV.

HOW VACCINES ARE MADE?

Vaccines are made using parts of a pathogen or weakened forms of the whole organism that can safely stimulate the immune system. These components act as **antigens**, triggering the body to produce antibodies and memory cells that provide protection against future infections.

MAJOR TYPES OF VACCINES

1. **Subunit Vaccines** – Contain specific proteins or lipids from a pathogen; can be produced using recombinant DNA technology (e.g., hepatitis B, Novavax COVID-19).
2. **Attenuated Vaccines** – Use live but weakened pathogens to produce strong and long-lasting immunity (e.g., MMR, Sabin polio).
3. **Inactivated (Killed) Vaccines** – Contain killed pathogens; often require booster shots (e.g., Salk polio, rabies, Sinovac COVID-19).

MAJOR TYPES OF VACCINES

4. **DNA-Based Vaccines** – Use plasmids carrying antigen genes that cause cells to produce the antigen (e.g., West Nile in horses, AstraZeneca COVID-19).
5. **mRNA Vaccines** – Use messenger RNA to instruct cells to produce a harmless viral protein that triggers immunity (e.g., Pfizer-BioNTech, Moderna).

RECOMBINANT SUBUNIT VACCINES

Before recombinant DNA technology, subunit vaccines were made by growing pathogens and purifying their surface proteins, which carried contamination risks. Today, scientists clone antigen genes into plasmids and insert them into microbes like *Escherichia coli* or yeast, allowing large amounts of the antigenic protein to be safely produced and purified for vaccines.

EXAMPLES OF RECOMBINANT VACCINES

- **Hepatitis B vaccine** – Yeast-produced hepatitis B surface antigen
- **Gardasil (HPV vaccine)** – Protects against HPV 16, 18 (cervical cancer) and 6, 11 (genital warts); three doses
- Shows how microbial biotechnology produces important vaccine proteins.

BOOSTER IMMUNIZATION

Some vaccines (especially inactivated types) do not provide lifelong protection. Antibody levels decline over time, and booster shots strengthen immune memory.

- **DPT and tetanus** – effective for about 10 years
- **Influenza vaccine** – requires annual immunization

"Map it out"

Rubrics

Criteria	Excellent (4 pts)	Good (3 pts)	Fair (2 pts)	Needs improvement (1 pt)
Content Accuracy	All information is scientifically accurate; no errors present.	Minor errors but does not affect overall understanding.	Some inaccuracies that affect clarity.	Many incorrect or misleading concepts.
Organization & Connections	Concepts are logically arranged; connections are clearly labeled and meaningful.	Mostly organized with clear connections.	Some connections unclear or poorly arranged.	Disorganized; relationships not clearly shown.
Understanding	Shows clear relationships between ideas and deeper explanation of processes.	Shows basic understanding of relationships.	Limited explanation; mostly listing ideas.	Minimal understanding shown.
Presentation	Neat, readable, and visually appealing; effective use of arrows, labels, and structure.	Generally neat and readable.	Somewhat messy but understandable.	Difficult to read or poorly presented.